

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## ASSESSMENT TOOL 3 – TECHNICAL PRESENTATION

|                  |                                                                                                                                                                |               |                                            |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------------------------|
| Course Code:     | CSA06                                                                                                                                                          | Course Title: | Design and Analysis of Algorithms          |
| CO Assessed:     | CO1 – Precision (BL1, BL2, BL3)                                                                                                                                | Assessment:   | Assessment Tool 3 – Technical Presentation |
| Weightage:       | 10% of CIA                                                                                                                                                     | Total Marks:  | 100 (1 Question × 100 Marks)               |
| CO1 Statement:   | <i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i> |               |                                            |
| Student Name:    | SHAFFRIA.A                                                                                                                                                     | Reg. No.:     | 192571047                                  |
| Section / Batch: | B TECH CSBS                                                                                                                                                    | Date:         | 25.07.2026                                 |

### Instructions to Students

- Answer the given question through a technical presentation. Total Marks: 100.
- Clearly define the problem and explain the concept of algorithm growth functions.
- Present comparative analysis using appropriate representations such as graphs, tables, or charts.
- Use suitable tools (PowerPoint, visualization tools, or software) to support your explanation.
- Ensure technical accuracy, logical flow, and proper interpretation of results.
- Maintain clarity in communication, proper slide organization, and professional delivery.
- Time management, confidence, and audience engagement will be considered during evaluation.

| Question Type          | Focus Area                                                                                                                              | Questions | Marks |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|
| Technical Presentation | Analyze and compare growth functions using mathematical techniques and asymptotic concepts and with graphical and visual interpretation | Q1        | 100   |
| GRAND TOTAL:           |                                                                                                                                         | 100 Marks |       |

### SECTION A – Technical Presentation

**Code of Conduct**

*I SHAFFRIA.A (192571047) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: \_\_\_\_\_

**RUBRICS FOR CALCULATION**

| Criteria                   | Wt. | Level 4 – Exemplary                                                | Level 3 – Proficient                                           | Level 2 – Developing                                       | Level 1 – Beginning                                                 |
|----------------------------|-----|--------------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------------|
| Technical Content Accuracy | 30  | Highly accurate, in-depth, and insightful technical explanation    | Technically correct content with adequate depth and clarity    | Basic concepts presented with limited accuracy and depth   | Content contains major technical errors; concepts poorly understood |
| Problem Identification     | 25  | Problem rigorously analyzed using appropriate and advanced methods | Problem clearly defined with a logical engineering approach    | Problem identified but analysis or methodology is weak     | Problem statement unclear or incorrect; no logical approach         |
| Use of Tools               | 20  | Advanced tools, well-analyzed data, and highly effective visuals   | Appropriate tools and data used effectively with clear visuals | Limited use of tools and basic data interpretation         | Minimal or incorrect use of tools and data; poor visuals            |
| Communication              | 15  | Highly engaging, confident, and audience-focused delivery          | Clear, organized, and professional communication               | Understandable but lacks clarity, structure, or confidence | Poor organization, unclear speech, ineffective slides               |
| Professionalism            | 10  | Exemplary professionalism, leadership, and ethical awareness       | Professional conduct with effective team collaboration         | Basic professionalism; uneven team participation           | Lacks professionalism; minimal contribution or ethical awareness    |

**Marks Evaluation:**

| Question No.        | Technical Content Accuracy (30%) | Problem Identification (25%) | Use of Tools (20%) | Communication (15%) | Professionalism (10%) | Total Marks (100) |
|---------------------|----------------------------------|------------------------------|--------------------|---------------------|-----------------------|-------------------|
| Q1                  |                                  |                              |                    |                     |                       |                   |
| Overall Total (100) |                                  |                              |                    |                     |                       |                   |

| Faculty In-charge | Course Coordinator | Head of Department |
|-------------------|--------------------|--------------------|
|                   |                    |                    |
| Signature: _____  | Signature: _____   | Signature: _____   |
| Date: _____       | Date: _____        | Date: _____        |

## **TECHNICAL PRESENTATION**

### **Q38. Efficiency Analysis of Polynomial Algorithms**

Analyze an algorithm with complexity  $O(n^3)$  and evaluate its scalability. Clearly define the problem and input size impact. Use graphical tools to show growth trend. Compare with lower-order complexities. Present conclusions on performance limitations.

